Accurate physical modeling and synchronization control of dual-linear-motor-driven gantry with dynamic load

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COLLECTIONS

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ABSTRACT

To achieve high-accuracy tracking of a dual-linear-motor-driven (DLMD) gantry, high-level synchronization between redundant actuators becomes a non-negligible factor and also a difficult issue to be solved priorly. Especially, when both X and Y axes are simultaneously operating to accomplish complex tasks efficiently, additional coupling effects will be generated by the dynamic load presented on the crossbeam, which makes the synchronization issue more complicated than the case with static load. However, due to the absence of an accurate model to fully reveal the complete coupling characteristics, existing approaches to this issue still have inherent limitations. Therefore, this paper focuses on the systematic physical modeling and synchronization control of a DLMD gantry with a dynamic load presented on the crossbeam. A complete coupling mathematical model is established first, by fully considering two linear motions (X-axis and Y-axis) and also including the additional rotational motion of the crossbeam. Built upon the effective model information, corresponding solutions by compensating the dynamic load effects and actively controlling the rotational dynamics to regulate the internal forces have been proposed, leading to a novel adaptive robust synchronization control method. The results of comparative experiments verify the effectiveness and superiority of the proposed method in dealing with the synchronization issue subjected to dynamic load effects.

I. INTRODUCTION

Among the various types of precision Cartesian robotic systems, the H-type gantry with a dual-driven structure and direct actuators[1–3] is widely used in industrial applications, in which a payload platform is driven by one motor along the crossbeam while two other parallel arranged motors are rigidly connected to drive the motion of the entire crossbeam. Although the dual-driven structure is used to provide larger thrust and achieve better control performance, more severe requirements of control also arise due to its mechanical coupling. Namely, the motions of two parallel arranged motors are naturally restricted by each other through the coupling bridge. Thus, except for the desired linear motion of this axis, the additional crossbeam rotation will result if the two motors are not synchronized well. Meanwhile, because of the physical restriction in the linear guide pairs, excessive internal forces might be generated simultaneously. Then, adverse effects, such as additional wear in the guiding parts and deformations or even damage of the hardware, may also result. Importantly, the normal operation of such kinds of systems may be affected, not to mention high-accuracy tracking. Especially, as a kind of typical multi-axis machines, both axes of the gantry system are usually commanded to operate together to accomplish complex tasks with high efficiency. However, the movement of the payload platform along the crossbeam will cause more complicated dynamic coupling effects between the motions of the two other axes (i.e., the dynamic load effects to be mainly studied in the following), which makes the realization of high-level synchronization more difficult.

To our knowledge, the synchronization control design on a dual-linear-motor-driven (DLMD) gantry is still far from mature. Part of the reasons might be due to the absence of an accurate dynamical model, leading to an incomplete knowledge of the system. Commonly, the motion of each motor is modeled individually,[4,5] while the mechanical coupling between two parallel motors is essentially ignored, which makes the resulting model actually have no guidance meaning on controller development. Other research studies on coupling effects can be mainly categorized into two categories. The first one is the coupling modeling of the entire system,[6–9] in which a three-degree-of-freedom (3DOF) modeling is usually presented, i.e., the linear motions of the X-axis and Y-axis and the rotational motion of the crossbeam. Although certain coupling effects can be explained by those models, inherent conflicts between the handing of physical constraints and the generation of rotational motion still exist, leading to doubts about the correctness of those works. For example, as in Refs. 8 and 9, under the assumption of pure rotational flexible joints, the rotation of the rigid crossbeam essentially cannot be generated. The second type is only focusing on the subsystem with a dual-driven structure.[10,11] By recognizing the relatively flexible feature of the bearing balls in direct-driven systems and subsequently modeling them by a set of distributed horizontal two-way springs,[12] the unbalanced deformations of those components along with additional internal forces can be clearly indicated. Then, the generation of additional rotational motion of the crossbeam and the aforementioned adverse effects are properly explained. However, the movement of the platform on the crossbeam has not been involved, which means the complex coupling between the motions of two axes and the dynamic load effects are not fully considered. Therefore, an accurate dynamical model to reveal the complete properties of such kinds of systems is still of great concern.

Limited by the aforementioned situation of modeling, earlier synchronized control schemes are usually based on the pure motion synchronization, e.g., the synchronized master command approach,[13] the master-slave control,[4] and the well-known cross-coupled control.[5,14–16] Due to the ignorance of mechanical coupling in dual-driven systems and the following internal force issue, the achievable synchronization performance is actually limited, which has been examined in detail in Ref. 10. Meanwhile, as in cross-coupled control, the design of the cross-coupled compensator and the set of the related controller gains have no systematic guidelines, especially for the dynamic load condition. As opposed to those control schemes based on pure motion synchronization, Li and coworkers developed a kind of thrust allocation technique based synchronization method,[10] by using the idea of internal force regulation. Comparative experiments with the previous methods indicate the superiority of the proposed method. Essentially, it provides a simplified static solution to balance the rotational moment of the crossbeam. Thus, the transient performance of internal force regulation is still limited. To overcome this weakness, a novel synchronization method by directly controlling the rotational dynamics to regulate the internal forces more efficiently has been proposed in Ref. 11. The synchronization performance is further improved, simultaneously indicating the relationship between internal forces and rotational angles. However, similar to most of the existing works, only the simple case with static load is studied. Actually, systematic work on dealing with the synchronization issue subjected to dynamic load effects can rarely been seen yet, leading to the urgent demand of further work.

On the other hand, advanced control technology is also indispensable for dealing with various uncertainties in practice, such as sliding mode control (SMC),[2,3] optimal control,[15] adaptive control (AC),[18–20] robust control (RC).[21,22] Likewise, a series of performance oriented adaptive robust control (ARC)[23–25] algorithms considering both parametric uncertainties and uncertain nonlinearities have been proposed. In addition, they have been successfully applied to direct-driven systems.[11,12,26,27] Meanwhile, an integrated direct/indirect ARC (DIARC) approach[25] is proposed to simultaneously obtain fast and accurate parameter estimation.

In this paper, to efficiently guide the development of the synchronization control scheme, an accurate physical modeling is presented first to capture the essential characteristics of DLMD gantry systems. Taking into account two obvious linear motions (X-axis and Y-axis) and the usually ignored rotation of the crossbeam along with the deformations of bearing balls, an integrated 3DOF dynamical model is derived to clearly reveal the essence of dynamic load effects. Built upon this knowledge, following the idea by actively suppressing the rotational dynamics to effectively regulate the internal forces, a novel three-input three-output (TITO) synchronization control method is then proposed. The dynamic load effects and various uncertainties are effectively compensated and dominated by the ARC controller. The proposed method is compared to the existing thrust allocation technique based synchronization methods. Experimental results on a practical DLMD gantry show the significant effect of dynamic load on the synchronization performance and also verify the performance improvement of our proposed synchronization strategy.

This paper is organized as follows: In Sec. II, a detailed system description, the modeling procedure, and the corresponding control objectives are presented. In Sec. III, a new synchronization scheme is proposed, and the corresponding adaptive robust synchronization controller is synthesized. In Sec. IV, the comparison experiments are designed and compared. Finally, a conclusion is drawn in Sec. V.

II. PROBLEM FORMULATION

A. System description

A general DLMD gantry with X- and Y-axis sub-assemblies is shown in Fig. 1. Commonly, to guarantee the repeated positioning

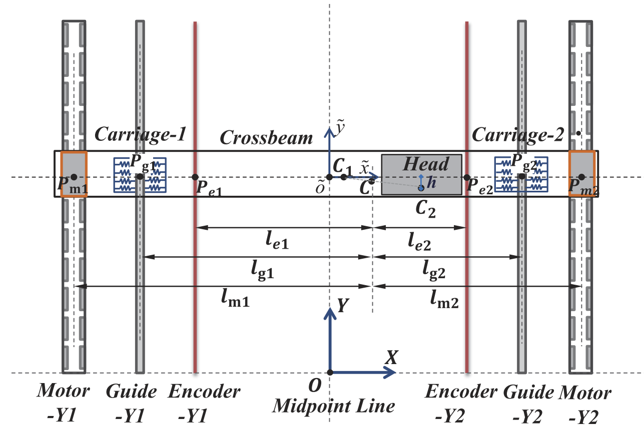

FIG. 1. General configuration of a dual-driven gantry.

accuracy through mechanical structure, a crossbeam of a large span and mass is rigidly connected with two parallel motors and sliders. The linear motion of the Y-axis is guided by two parallel pairs of ball linear guide rails. Thus, the motions of two parallel motors of the Y-axis are not mutually independent but naturally restricted by each other through the coupling bridge. Meanwhile, a payload platform, i.e., the working head, is driven by another linear motor along the crossbeam to provide the linear motion of the X-axis. Despite the high-rigidity achieved by such kinds of structures, certain components of relative flexibility still exist due to various factors, e.g., for the successful assembling in practice. According to the existing research,[10–12] the relative elastic property of bearing balls is most significant comparing to the other rigid components and connections. Thus, they can be modeled as distributed horizontal two-way springs with large stiffness.

Within this structure, it is usually assumed that the parallelism of two linear guides can be sufficiently guaranteed owing to the high-accuracy assembling in practice. When the crossbeam is orthogonal to the parallel guides of the Y-axis, the horizontal two-way springs can be taken as perfectly balanced, as shown in Fig. 1, which is defined as the equilibrium state without additional internal forces in this paper. However, subjected to the physical restriction between the carriages and the guide rails, the bearing balls would be squeezed unevenly when the thrusts of two coupled actuators are not properly allocated, leading to the unbalanced lateral deformations, as shown in Fig. 2. Because of the high stiffness of the bearing balls, excessive internal forces could be exhibited, which will cause some adverse effects such as increase in mechanical wear, influence on normal operation, and even hardware damage. Meanwhile, the contemporaneous rotation of the crossbeam (i.e., the "pull and drag" phenomena) will also decrease the feed-accuracy of the end-effector, which is usually attached to the working head in practice.

Furthermore, complex coupling effects exist between the working head's linear motion and the crossbeam's rotation. Meanwhile, the physical property of the Y-axis will be changed due to the dynamic load presented on the crossbeam. In short, denote the mass centers of the crossbeam, the working head, and the entire moving body of the Y-axis as C_1, C_2, and C, respectively, as shown in Fig. 1. Define the spans of the two motors, encoders, and linear guides, as shown in Fig. 1, as l_g, l_e, and l_m, respectively, with $|CP_{\bullet i}| = l_{\bullet i}$ being the distance from C to the corresponding object. Obviously, the position of C varies with the movement of the working head, i.e., varying $l_{\bullet i}$. Thus, it is particularly difficult to achieve high-level synchronization of such kinds of systems with dynamic load, not to mention high-accuracy tracking performance.

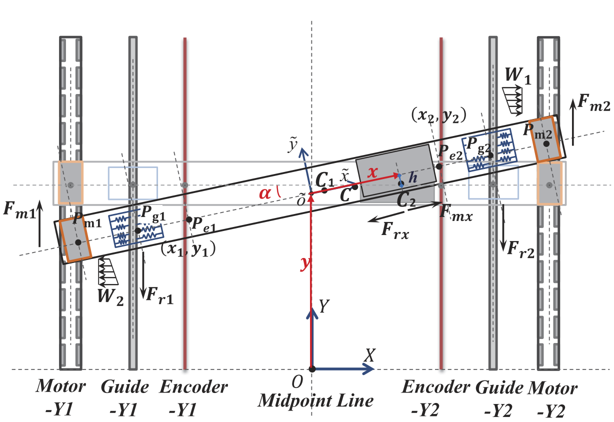

FIG. 2. Linear and rotational motions of the rigid moving part.

In general, high-accuracy tracking is usually the common requirement of precision systems. However, as analyzed previously, high-level synchronization is the basis to achieve the former goal for DLMD gantry systems. As a result, both high-level synchronization between redundant actuators and high-accuracy tracking should be concurrently considered. Therefore, both objectives are pursued in this paper, where the synchronization issue is mainly discussed.

B. Rigid-flexible modeling

To deal with the complicated synchronization issue with dynamic load, an accurate dynamical model exactly revealing the dynamical properties is a prior requirement to develop effective control methods. From the DLMD gantry shown in Fig. 2, the complete motions can be represented by the traditional rigid linear motions of X and Y axes along the corresponding linear guides (i.e., represented by x and y) and the usually ignored flexible rotation of the crossbeam (i.e., represented by α). Then, the rigid-flexible modeling can be performed quickly as follows:

Denote the mass of the crossbeam and the working head as M_b and M_x, respectively. Meanwhile, denote the moment of inertia of the crossbeam with respect to (w.r.t.) $\tilde{o}$ and the working head w.r.t. C_2 as J_{M_b} and J_{M_x}, respectively. The positions of M_b and M_x in the world frame OXY can be represented by

$$\begin{aligned} x_{M_b} &= 0, & x_{M_x} &= x\cos\alpha + h\sin\alpha, \\ y_{M_b} &= y, & y_{M_x} &= y - h\cos\alpha + x\sin\alpha, \end{aligned} \tag{1}$$

which lead to the following velocity vector

$$v_{M_b} = \begin{bmatrix} 0 \\ \dot{y} \end{bmatrix}, \quad v_{M_x} = \begin{bmatrix} (\dot{x} + h\dot{\alpha})\cos\alpha - x\dot{\alpha}\sin\alpha \\ \dot{y} + (h\dot{\alpha} + \dot{x})\sin\alpha + x\dot{\alpha}\cos\alpha \end{bmatrix}. \tag{2}$$

Thus, the entire kinetic energy can be calculated as

$$\begin{aligned} E_k &= E_{kb} + E_{kx} \\ &= \frac{1}{2}M_b v_{M_b}^T v_{M_b} + \frac{1}{2}J_{M_b}\dot{\alpha}^2 + \frac{1}{2}M_x v_{M_x}^T v_{M_x} + \frac{1}{2}J_{M_x}\dot{\alpha}^2 \\ &= \frac{1}{2}M_y\dot{y}^2 + \frac{1}{2}M_x\dot{x}^2 + \frac{1}{2}\left[J_M + M_x(x^2 + h^2)\right]\dot{\alpha}^2 \\ &\quad + \dot{x}\dot{y}M_x\sin\alpha + \dot{x}\dot{\alpha}M_x h + \dot{y}\dot{\alpha}M_x(x\cos\alpha + h\sin\alpha), \end{aligned} \tag{3}$$

where $M_y = M_b + M_x$ and $J_M = J_{M_b} + J_{M_x}$. Noting that due to the flexible deformations of bearing balls, the following potential energy will also be generated:

$$E_p = \frac{1}{2}K_\alpha\alpha^2, \tag{4}$$

where K_α is the equivalent rotational stiffness of the bearing balls modeled in Refs. 10 and 12, indicating the direct relationship between additional internal forces and rotational angles. Combined (4) with (3), the total energy of the system can be obtained by E

$= E_k - E_p$. By applying the Lagrangian method, i.e., $\frac{d}{dt}\left(\frac{\partial E}{\partial \dot{q}_i}\right) - \frac{\partial E}{\partial q_i} = F_i, i = 1, 2, 3$, one can obtain

$$M_q\ddot{q} + C_q\dot{q} + K_q q = F, \tag{5}$$

where $q = [x,\ y,\ \alpha]$ is the set of generalized coordinates; M_q, C_q, and K_q are the matrices of inertia, Coriolis and centrifugal forces, and stiffness, respectively; and F represents the matrix of generalized forces to be calculated later. As can be seen from Fig. 2, without direct analysis of the unknown disturbance forces, the thrust F_{m*} of each motor and the friction force F_{r*} of each guide pair are the main external forces. Note that linear motors of the permanent magnet synchronous type with a kHz bandwidth of the current loop are used in our research. Thus, by neglecting the fast electrical dynamics, a popular simplified model of F_{m*} will be used throughout this paper. Therefore, the external forces can be described by

$$F_{mx} = K_x u_x, \quad F_{m1} = K_1 u_1, \quad F_{m2} = K_2 u_2, \tag{6}$$

$$\begin{gathered} F_{rx} = B_x\dot{x} + F_{fx}(\dot{x}), \\ F_{r1} = B_1\dot{y}_{g1} + F_{f1}(\dot{y}_{g1}), \\ F_{r2} = B_2\dot{y}_{g2} + F_{f2}(\dot{y}_{g2}), \end{gathered} \tag{7}$$

where u_* represents the control input of each motor, with K_* being the corresponding force constant; B_* denotes the coefficient of viscosity; $\dot{y}_{gi}$ denotes the velocity of carriage-i, as shown in Fig. 2; and the Coulomb friction F_{f*} can be approximated by a simple smooth function given by $\bar{F}_{f*}(\bullet) = A_{f*}S_f(\bullet)$, with A_{f*} being the coefficient of Coulomb friction and $S_f(\bullet) = \frac{2}{\pi}atan(1000\bullet)$ being the function used to approximate $s\,\mathrm{gn}(\bullet)$. According to the analysis of virtual displacement given in Table I and the principle of virtual work, the detailed forms of generalized forces can be derived,

$$F = \begin{bmatrix} F_{mx} - F_{rx} \\ F_{m1} + F_{m2} - F_{r1} - F_{r2} + (F_{mx} - F_{rx})\sin\alpha \\ (F_{m2} - F_{m1})\frac{l_m}{2} + (F_{r1} - F_{r2})\frac{l_g}{2} + (F_{mx} - F_{rx})h \end{bmatrix}. \tag{8}$$

Thus, by recognizing the geometrical relationship between y_{gi}, y, and α, i.e., $y_{gi} = y + \frac{l_g}{2}\alpha(-1)^i,\ \ i = 1, 2$, the following standard dynamic form w.r.t. q can be obtained:

$$M_q\ddot{q} + C_q\dot{q} + B_q\dot{q} + A_q S_f(\dot{q}) + K_q q = K_1 T u_q + d, \tag{9}$$

where $u_q = [u_x, u_1, u_2]^T$; $S_f(\dot{q}) = [S_f(\dot{x}), S_f(\dot{y}), S_f(\dot{\alpha})]^T$; and $d = [d_x,\ d_y,\ d_\alpha]$ represents the lumped modeling errors, including the approximation error of Coulomb friction and the previously ignored unknown disturbances.

TABLE I. Virtual displacement calculation.

Force	Virtual displacement
F_{m1}	$\delta y - \frac{l_m}{2}\delta\alpha$
F_{r1}	$\delta y - \frac{l_g}{2}\delta\alpha$
F_{m2}	$\delta y + \frac{l_m}{2}\delta\alpha$
F_{r2}	$\delta y + \frac{l_g}{2}\delta\alpha$
$(F_{mx} - F_{rx})\cos\alpha$	$\cos\alpha\delta x - x\sin\alpha\delta\alpha + h\cos\alpha\delta\alpha$
$(F_{mx} - F_{rx})\sin\alpha$	$\delta y + \sin\alpha\delta x + h\sin\alpha\delta\alpha + x\cos\alpha\delta\alpha$

For simplicity, normalizing (9) w.r.t. the force constant K_1, one has

$$M_k\ddot{q} + C_k\dot{q} + B_k\dot{q} + A_k S_f(\dot{q}) + K_k q = v + d_{kn} + \tilde{d}_k, v = T u_q, \tag{10}$$

$$\begin{gathered} M_k = \begin{bmatrix} \bar{M}_x & \bar{M}_x\sin\alpha & \bar{M}_x h \\ \bar{M}_x\sin\alpha & \bar{M}_y & \bar{M}_x(h\sin\alpha + x\cos\alpha) \\ \bar{M}_x h & \bar{M}_x(h\sin\alpha + x\cos\alpha) & \bar{J}_M + \bar{M}_x(h^2 + x^2) \end{bmatrix}, \\ C_k = \begin{bmatrix} 0 & 0 & -\bar{M}_x x\dot{\alpha} \\ \bar{M}_x\dot{\alpha}\cos\alpha & 0 & \bar{M}_x[(h\dot{\alpha} + \dot{x})\cos\alpha - x\dot{\alpha}\sin\alpha] \\ \bar{M}_x x\dot{\alpha} & 0 & \bar{M}_x x\dot{x} \end{bmatrix}, \\ B_k = \begin{bmatrix} \bar{B}_x & 0 & 0 \\ \bar{B}_x\sin\alpha & \bar{B}_y & \bar{B}_c\frac{l_g}{2} \\ h\bar{B}_x & \bar{B}_c\frac{l_g}{2} & \bar{B}_y\frac{l_g^2}{4} \end{bmatrix}, \quad A = \begin{bmatrix} \bar{A}_{fx} & 0 & 0 \\ \bar{A}_{fx}\sin\alpha & \bar{A}_{fy} & 0 \\ h\bar{A}_{fx} & \bar{A}_{fc}\frac{l_g}{2} & 0 \end{bmatrix}, \\ K_k = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \bar{K}_\alpha \end{bmatrix}, \quad T = \begin{bmatrix} k_{mx} & 0 & 0 \\ k_{mx}\sin\alpha & 1 & 1 \\ k_{mx}h & -\frac{l_m}{2} & k_{my}\frac{l_m}{2} \end{bmatrix}, \end{gathered} \tag{11}$$

where $\bullet_k$ represents the normalized matrix and d_{kn} denotes the constant component of d_k, with $\tilde{d}_k = d_k - d_{kn}$. The detailed forms of those coefficient matrices are given in (11), with $k_{mx} = K_x/K_1$, $k_{my} = K_2/K_1$, $\bullet_y = \bullet_1 + \bullet_2$, $\bullet_c = \bullet_2 - \bullet_1$, and $\bar{*}$ being the normalized parameters, i.e., $\bar{*} = \frac{*}{K_1}$. Thus, the complicated coupling effects between the motions of two axes can be clearly indicated. For example, the dynamic load effects can be mainly described by the varying inertia term $\bar{J}_M + \bar{M}_x(h^2 + x^2)$ and the inertia moment term $\bar{M}_x(h\sin\alpha + x\cos\alpha)\ddot{y}$, where $\bar{M}_x\ddot{y}$ is the inertia force of the working head along the Y-axis and $(h\sin\alpha + x\cos\alpha)$ is the active length.

Apparently, the following properties always hold good for dynamics (10): (**P1**) In any finite work space Ω_q, M_k is a symmetric positive definite (s.p.d.) matrix satisfying $\varsigma_1 I \le M_k \le \varsigma_2 I$, with ς_1 and ς_2 being positive scalars; (**P2**) $N_k = \dot{M}_k - 2C_k$ is skew-symmetric, i.e., $s^T N_k s = 0, \forall s$; (**P3**) The matrices M_k, C_k, B_k, A_k, and d_{kn} can be linearly parameterized by the following set θ of unknown parameters,

$$\begin{aligned} \theta &= [\theta_1, \ldots, \theta_{13}]^T \\ &= \left[\bar{M}_x, \bar{B}_x, \bar{A}_{fx}, \bar{d}_{xn}, \bar{M}_y, \bar{B}_y, \bar{A}_{fy}, \bar{d}_{yn}, \bar{J}_M, \frac{\bar{K}_\alpha}{1e4}, \bar{B}_c, \bar{A}_{fc}, \bar{d}_{\alpha n}\right]^T. \end{aligned} \tag{12}$$

Furthermore, for simplicity, the definitions of the general symbols used in the following are summarized here: $\hat{\bullet}$ and $\tilde{\bullet}$ denote the estimate and estimation error of $\bullet$, i.e., $\tilde{\bullet} = \hat{\bullet} - \bullet$; $\sigma_{\max}(\bullet)$ and $\sigma_{\min}(\bullet)$ represent the maximum and minimum eigenvalue of $\bullet$; $\bullet_{max}$ and $\bullet_{min}$ are the maximum and minimum value of $\bullet(t)$ for all t, respectively.

C. Assumption and control objective

Generally, various uncertainties, like parametric variations and uncertain nonlinearities, always exist in practice. However, it is also a fact that all those uncertainties are within certain finite bounds. Thus, the following assumption can be made first to account for those practical situations.

Assumption 1. *The parametric variations and uncertain nonlinearities have known bounds, i.e.,*

$$\theta \in \Omega_\theta \triangleq \{\theta : \theta_{\min} \le \theta \le \theta_{\max}\} \tag{13}$$

$$\tilde{d} \in \Omega_d \triangleq \{\tilde{d} : \|\tilde{d}\| \le \delta_d\}, \tag{14}$$

where $\theta_{\min} = [\theta_{1\min}, \ldots, \theta_{13\min}]^T$ and $\theta_{\max} = [\theta_{1\max}, \ldots, \theta_{13\max}]^T$ *are the lower and upper bounds of* θ*, respectively, and* δ_d *is a certain bounding function.*

Therefore, combined with the basic requirements of high-accuracy tracking of both axes and high-level synchronization with dynamic load presented on the crossbeam for DLMD gantry systems, the detailed control objectives in this work are summarized as follows: Subjected to the aforementioned uncertainties and the dynamic load effects emphatically discussed in this paper, a proper control input u_q is to be synthesized such that the linear motions of both axes $x(t)$ and $y(t)$ can track the corresponding smooth reference trajectories $x_d(t)$ and $y_d(t)$ as closely as possible, and simultaneously, the rotational motion $\alpha(t)$ has to be highly suppressed to avoid large internal forces during both transient and steady-state periods. The following error index is defined as

$$e = [e_x, e_y, e_\alpha]^T = [x(t) - x_d(t), y(t) - y_d(t), \alpha(t)]^T. \tag{15}$$

Thus, by making e small, the above-mentioned objectives can be achieved.

III. ADAPTIVE ROBUST SYNCHRONIZATION CONTROLLER

In this section, a control algorithm is presented to achieve high-level synchronization and high-accuracy tracking performance of DLMD gantry systems. The desired compensation technique is applied to form desired compensation integrated direct/indirect adaptive robust control (DCDIARC). The overall block diagram of the control design is shown in Fig. 3. With the use of the previously established dynamical model, the unknown parameters will be accurately estimated online to properly compensate the main coupling effects and the parametric uncertainties, while the remaining uncertainties will be dominated by the robust feedback in this framework. The detailed design procedure is as follows:

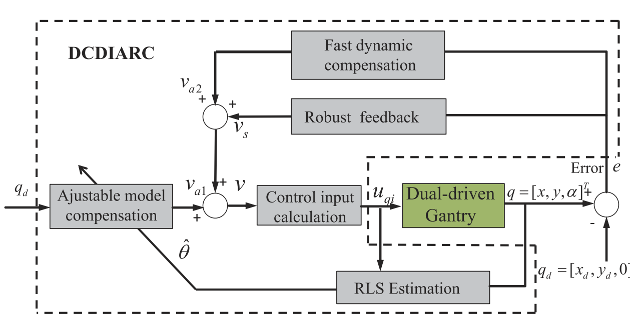

FIG. 3. Block diagram of the closed-loop system.

A. Adaptation law

To separate the designs of the controller and the estimator for different purposes, i.e., robust tracking and accurate parameter estimation, the following rate-limited projection structure is used for online adaptation:

$$\dot{\hat{\theta}} = sat_{\dot{\hat{\theta}}_M}(Proj_{\hat{\theta}}(\Gamma\tau)), \tag{16}$$

where Γ is the adaptation rate matrix; $\dot{\hat{\theta}}_M$ is the preset upper bound for the adaptation rate; τ is the adaptation function; the detailed forms of Γ and τ will be synthesized later; $sat_{\dot{\hat{\theta}}_M}$ is the saturation function of the following form,

$$sat_{\dot{\hat{\theta}}_M}(\bullet) = s_0\bullet, s_0 = \begin{cases} 1, & if\,\|\bullet\| \le \dot{\hat{\theta}}_M \\ \frac{\dot{\hat{\theta}}_M}{\|\bullet\|}, & if\,\|\bullet\| > \dot{\hat{\theta}}_M, \end{cases} \tag{17}$$

and the projection mapping $Proj_{\hat{\theta}}$ is given by

$$Proj_{\hat{\theta}}(\bullet) = \begin{cases} \bullet, & \hat{\theta} \in \bar{\Omega}_\theta \; or \; n_{\hat{\theta}}^T\bullet < 0 \\ \left(I - \Gamma\frac{n_{\hat{\theta}} n_{\hat{\theta}}^T}{n_{\hat{\theta}}^T \Gamma n_{\hat{\theta}}}\right)\bullet, & \hat{\theta} \in \partial\Omega_\theta \,\&\, n_{\hat{\theta}}^T\bullet > 0, \end{cases} \tag{18}$$

where $\partial\Omega_\theta$ and $\bar{\Omega}_\theta$ denote the boundary and the interior of the closed set Ω_θ; $n_{\hat{\theta}_\theta}$ represents the unit normal vector on the boundary in the outward direction.

As seen from (17) and (18), it is obvious that the adaptation law (16) have the following properties for any τ:[25] **(P4)** $\theta_{jmin} \le \hat{\theta}_j \le \theta_{jmax}$ for $j = 1, \ldots, 13$, **(P5)** $\tilde{\theta}^T(\Gamma^{-1}Proj_{\hat{\theta}}(\Gamma\tau) - \tau) \le 0, \forall\tau$, and **(P6)** $\|\dot{\hat{\theta}}\| \le \dot{\theta}_M$.

B. DCDIARC

As for the controller, the following quantity is defined first to simplify the design procedure:

$$s = \dot{e} + \Lambda e, \tag{19}$$

where Λ is a 3×3 positive definite (p.d.) diagonal matrix.

Then, a positive semidefinite (p.s.d.) Lyapunov function is chosen,

$$V(t) = \frac{1}{2}s^T M_k s + \frac{1}{2}e^T K_e e, \tag{20}$$

where K_e is a 3×3 p.d. matrix. Differentiating $V(t)$ and noting **(P2)** to eliminate the term $\frac{1}{2}s^T\dot{M}_k s$, one can obtain

$$\begin{aligned} \dot{V}(t) &= s^T\big[v + d_{kn} - M_k\ddot{q}_d - C_k\dot{q}_d - B_k\dot{q}_d - K_k q_d - A_k S_f(\dot{q}_d) \\ &\quad - B_k\dot{e} - K_k e - A_k g_k \dot{e} + C_k\Lambda e + M_k\Lambda\dot{e} + \tilde{d}_k\big] + e^T K_e \dot{e} \\ &= s^T[v + \Psi(q,\dot{q},t)\theta + \tilde{d}_k] + e^T K_e \dot{e}, \end{aligned} \tag{21}$$

where $q_d(t) = [x_d(t), y_d(t), 0]^T$ represents the desired trajectory; $g_k(\dot{q},t)$ is a 3×3 matrix consisting of known functions to bind the approximation error, i.e., $g_k(\dot{q},t)\dot{e} = S_f(\dot{q}) - S_f(\dot{q}_d)$; and Ψ is a 3×13 regressor matrix. Thus, partly seen from (21), $\Psi(q,\dot{q},t)$ can be easily divided into two parts,

$$\Psi(q,\dot{q},t) = \tilde{\Psi}(e,\dot{e},t) + \Psi_d(q_d,\dot{q}_d,t), \tag{22}$$

where $\Psi_d(q_d, \dot{q}_d, t)$ is the desired part containing q_d only and $\tilde{\Psi}(e, \dot{e}, t)$ includes the remaining parts. Furthermore, according to Ref. 28, $\tilde{\Psi}(e, \dot{e}, t)$ can be qualified by the following terms:

$$\|\tilde{\Psi}\theta\| \le \zeta_1\|e\| + \zeta_2\|e\|^2 + \zeta_3\|s\| + \zeta_4\|s\|\|e\|, \tag{23}$$

where ζ_i, $i = 1, \ldots, 4$ is a positive constant.

Then, applying the desired compensation technique to reduce the influence of measurement noise, i.e., the desired regressor Ψ_d is used for model compensation, the resulting DCDIARC law has the following form:

$$v = v_a + v_s, v_a = v_{a1} + v_{a2}, v_{a1} = -\Psi_d\hat{\theta}, \tag{24}$$

where v_{a1} is the adjustable model compensation used to deal with the main influences of dynamic load, e.g., $\bar{M}_x(h\sin\alpha + x\cos\alpha)\ddot{y}$, and also for the purpose of perfect tracking; v_{a2} is the additional term to be synthesized later to obtain fast dynamic compensation; and v_s is the robust control term to guarantee the robust performance.

To account for (23), the robust term v_s is designed as

$$v_s = v_{s1} + v_{s2};\quad v_{s1} = -K_{s1}s - K_e e - K_a\|e\|^2 s, \tag{25}$$

where v_{s1} and v_{s2} are two components to be used for different purposes; K, K_e, and K_a are all 3×3 p.d. controller gain matrices designed as follows:

$$\begin{aligned}\sigma_{min}(K_a) \ge \zeta_2 + \zeta_4,\quad \sigma_{min}(K_e\Lambda) \ge \frac{1}{2}\zeta_1 + \frac{1}{4}\zeta_2,\\ \sigma_{min}(K_{s1}) \ge \frac{1}{2}\zeta_1 + \zeta_3 + \frac{1}{4}\zeta_4\end{aligned} \tag{26}$$

to satisfy the following condition:

$$Q = \begin{bmatrix}\sigma_{\min}(K_e\Lambda) - \frac{1}{4}\zeta_2 & -\frac{1}{2}\zeta_1\\ -\frac{1}{2}\zeta_1 & \sigma_{\min}(K) - \zeta_3 - \frac{1}{4}\zeta_4\end{bmatrix} > 0. \tag{27}$$

Then, v_{a2} and v_{s2} are designed to deal with the remaining lumped model uncertainties in (21), which are defined as

$$d_c + d^* = -\Psi_d\tilde{\theta} + \tilde{d}_k, \tag{28}$$

where d_c and d^* represent the different components of $-\Psi_d\tilde{\theta} + \tilde{d}_k$ in the low frequency range and high frequency range. Thus, the low frequency component d_c can be mainly compensated by v_{a2} with the following form:

$$v_{a2} = -\hat{d}_c, \dot{\hat{d}}_c = Proj_{\hat{d}_c}(\gamma_d s), \left|\hat{d}_c\right| \le d_{c\max}, \tag{29}$$

where γ_d is a 3×3 p.d. diagonal matrix and $d_{c\max}$ is a preset bound. Meanwhile, v_{s2} is designed to satisfy the following conditions:

$$\begin{aligned}&(i)\; s^T\{v_{s2} - \tilde{d}_c + d^*\} \le \eta,\\ &(ii)\qquad s^T v_{s2} \le 0,\end{aligned} \tag{30}$$

where $\eta = \eta_1\delta^2 + \eta_2\|\tilde{\theta}\|^2 + \eta_3 d_{c\max}{}^2$, with η_i being positive constants to dominate uncertainties. For example, v_{s2} having the following form can satisfy the above two conditions:

$$v_{s2} = -K_{s2}s,\quad K_{s2} \ge (c_1 + c_2\|\Psi_d\|^2 + c_3), \tag{31}$$

where c_i is a positive constant, with $c_i = \frac{1}{4\eta_i}$, $i = 1, \ldots, 3$.

Theorem 1. *The proposed DCDIARC control law (24)* and *(25) with the projection adaptation law (16) can always guarantee the boundedness of the control inputs and states of the closed-loop system. Furthermore, the Lyapunov function $V(t)$ in (20) has the following upper bound:*

$$V(t) \le \exp(-\varsigma t)V(0) + \frac{\eta}{\varsigma}[1 - \exp(-\varsigma t)], \tag{32}$$

where $\varsigma = 2\sigma_{\min}(Q)/\{\max[\sigma_{\max}(M_k), \sigma_{\max}(K_e)]\}$. Thus, the tracking errors are guaranteed to achieve prescribed transient and steady-state performances.

Proof 1. Substituting (19), (22), (24), and (29) into (21), and noting the definition in (28), yields

$$\begin{aligned}\dot{V}(t) &= s^T[-\Psi_d\hat{\theta} - \hat{d}_c + v_s + (\tilde{\Psi} + \Psi_d)\theta + \tilde{d}_k] + e^T K_e\dot{e}\\ &= s^T[v_s - \hat{d}_c + (-\Psi_d\tilde{\theta} + \tilde{d}_k) + \tilde{\Psi}\theta] + e^T K_e\dot{e}\\ &= s^T(v_s - \tilde{d}_c + d^* + \tilde{\Psi}\theta) + e^T K_e(s - \Lambda e)\\ &\le s^T(v_s - \tilde{d}_c + d^*) + \|s\|\|\tilde{\Psi}\theta\| + e^T K_e s - e^T K_e\Lambda e.\end{aligned} \tag{33}$$

Then, substituting (23), (25), and (26) into (33), we have

$$\begin{aligned}\dot{V}(t) \le& -\sigma_{\min}(K_{s1})\|s\|^2 - (\zeta_2 + \zeta_4)\|e\|^2\|s\|^2 + \zeta_1\|s\|\|e\|\\ &+ \zeta_2\|s\|\|e\|^2 + \zeta_3\|s\|^2 + \zeta_4\|s\|^2\|e\| - \sigma_{\min}(K_e\Lambda)\|e\|^2\\ &+ s^T(v_{s2} - \tilde{d}_c + d_c{}^*)\\ \le& -\zeta_2\|e\|^2\left(\|s\| - \frac{1}{2}\right)^2 - \zeta_4\|s\|^2\left(\|e\| - \frac{1}{2}\right)^2\\ &- \left[\sigma_{\min}(K_{s1}) - \zeta_3 - \frac{1}{4}\zeta_4\right]\|s\|^2 + \zeta_1\|s\|\|e\|\\ &- \left[\sigma_{\min}(K_e\Lambda) - \frac{1}{4}\zeta_2\right]\|e\|^2 + s^T(v_{s2} - \tilde{d}_c + d_c{}^*)\\ \le& -\left[\sigma_{\min}(K_{s1}) - \zeta_3 - \frac{1}{4}\zeta_4\right]\|s\|^2 + \zeta_1\|s\|\|e\|\\ &- \left[\sigma_{\min}(K_e\Lambda) - \frac{1}{4}\zeta_2\right]\|e\|^2 + s^T(v_{s2} - \tilde{d}_c + d_c{}^*).\end{aligned} \tag{34}$$

Noting (27) and condition i of *(30), (34) is equivalent to the following form:*

$$\dot{V}(t) \le -\chi^T Q\chi + \eta \le -\sigma_{\min}(Q)\chi^T\chi + \eta \le -\varsigma V + \eta, \tag{35}$$

with $\chi^T = [\|e\|\|s\|]$. Then, by comparison lemma, it is easy to derive (32) from (35), which proves Theorem 1.

C. Online parameter estimation

This subsection focuses on the fast and accurate parameter estimation, which may lead to better model compensation in the control design to further improve the performance. Especially, asymptotic tracking in the presence of parametric uncertainties only can be pursued here. By assuming $\tilde{d}_k = 0$ for the time being, (10) can be written in the following regression form:

$$v = \varphi^T\theta, \tag{36}$$

where φ^T is the corresponding regressor. By applying a stable filter $H_f(s)$ [e.g., $H_f(s) = 1/(\tau_f s+1)^2$] to both sides of (36), the following prediction error can be defined based on the resulting filtered dynamics:

$$\varepsilon = \varphi_f^T\hat{\theta} - \nu_f = \varphi_f^T\tilde{\theta}, \tag{37}$$

where $\bullet_f$ represents the filtered value of $\bullet$.

Then, for the purpose of fast and accurate online adaptation, the recursive least square (RLS) estimation algorithm[29] with both covariance re-setting and exponential forgetting techniques is used to design the estimator. Within the adaptation law (16), the resulting $\Gamma(t)$ and τ have the following forms:

$$\dot{\Gamma} = \begin{cases} \mu\Gamma - \frac{\Gamma\varphi_f\varphi_f^T\Gamma}{1+\kappa tr\{\varphi_f{}^T\Gamma\varphi_f\}} \; if & \sigma_{\max}(\Gamma(t)) \le \rho_M \\ 0 & otherwise, \end{cases} \tag{38}$$

$$\tau = -\frac{\varphi_f\varepsilon}{1+\kappa tr\{\varphi_f^T\Gamma\varphi_f\}}, \tag{39}$$

where $\mu \ge 0$ and $\kappa \ge 0$ denote the forgetting factor and the normalizing factor, respectively, and ρ_M is the preset upper bound for $\Gamma(t)$ to prevent the unstable situation of online estimation, i.e., $\sigma_{\max}(\Gamma(t)) \to \infty$, and always guarantees that $\Gamma(t) \le \rho_M I$.

Then, a similar lemma summarizing the properties of the estimator as stated in Ref. 30, and certain additional results including asymptotic tracking, can be further clarified:

Lemma 1. Within the rate-limited projection type adaptation framework (16), the RLS estimator given by (38) and *(39) guarantees the following:*

(a): *$\tilde{\theta}$, ε, and $\dot{\hat{\theta}}$ are uniformly bounded.*

(b): *ε and $\dot{\hat{\theta}}$ belong to L_2.*

Theorem 2. *If the system is void of uncertain nonlinearities after a finite time t_0, (i.e., $\tilde{d}_k = 0, \forall t \ge t_0$), once the following persistent excitation (PE) condition is satisfied,*

$$\int_t^{t+T} \varphi_f\varphi_f^T \ge \xi I, \text{ for some } \xi > 0 \text{ and } T > 0, \tag{40}$$

*and then, in addition to the robust performance given in **Theorem 1**, the control law (24)* and *(25) combined with the RLS estimator (38)* and *(39) can further guarantee that all the parameter estimates in $\hat{\theta}$ converge to their true values (i.e., $\tilde{\theta} \to 0$, as $t \to \infty$) and an improved final tracking – accuracy-asymptotic tracking – is also achieved (i.e., $e \to 0$ and $s \to 0$ as $t \to \infty$).*

Proof 2. From Lemma 1, since ε is uniformly bounded and $\varepsilon \in L_2$, by Barbalat's lemma, $\varepsilon \to 0$ as $t \to \infty$. Then, by noting (37), as $t \to \infty$, we have

$$\int_t^{t+T} \varepsilon^T(v)\varepsilon(v)dv = \int_t^{t+T} \tilde{\theta}^T(v)\varphi_f(v)\varphi_f^T(v)\tilde{\theta}(v)dv \to 0. \tag{41}$$

Similarly, from Lemma 1, $\dot{\hat{\theta}} \to 0$ as $t \to \infty$, i.e., *$\tilde{\theta}$ is constant as $t \to \infty$. Combing (40), one can obtain*

$$\tilde{\theta}^T \int_t^{t+T} \varphi_f(v)\varphi_f^T(v)dv\tilde{\theta} \ge \xi\|\tilde{\theta}\|^2 \to 0, \tag{42}$$

which obviously leads to $\tilde{\theta} \to 0$ as $t \to \infty$ and $\tilde{\theta} \in L_2$.

Then, another positive function is chosen as

$$V_a = V + \frac{1}{2}\hat{d}_c^T\gamma_d{}^{-1}\hat{d}_c. \tag{43}$$

Differentiating V_a and combining $\dot{V}_a$ with (27) and (34) yield

$$\dot{V}_a \le -\chi^TQ\chi + s^T(v_{s2} - \tilde{d}_c + d_c{}^*) + \hat{d}_c^T\gamma_d{}^{-1}\dot{\hat{d}}_c. \tag{44}$$

Noting **P5**, *(28), and condition (ii)* of *(30), (43) equals to*

$$\begin{aligned}\dot{V}_a &\le -\chi^TQ\chi + s^Tv_{s2} + s^T(-\Psi_d\tilde{\theta} - \hat{d}_c) + \hat{d}_c^T\gamma_d{}^{-1}\dot{\hat{d}}_c \\ &= -\chi^TQ\chi - s^T\Psi_d\tilde{\theta} + \hat{d}_c^T(-s + \gamma_d{}^{-1}\dot{\hat{d}}_c) \\ &\le -\chi^TQ\chi - s^T\Psi_d\tilde{\theta}.\end{aligned} \tag{45}$$

From Theorem 1, *all the signals are bounded,* i.e., *Ψ_d is bounded. Since $\tilde{\theta} \in L_2$, $\Psi_d\tilde{\theta} \in L_2$. Then, from inequality (44), it is easy to derive that $s \in L_2$ and $\chi \in L_2$. By Barbalat's lemma, $e \to 0$ as $t \to \infty$,* i.e., *asymptotic tracking is achieved.*

IV. EXPERIMENTAL RESULTS

A. Experimental setup

As shown in Fig. 4, comparative experiments are taken on a DLMD gantry set up in Zhejiang University. The geometric parameters of this stage are $l_m = 1.46\ m$, $l_e = 1.05\ m$, and $l_g = 1.28\ m$. Meanwhile, due to the almost symmetrical structure of this system, $|\overrightarrow{\bar{o}C_1}| \approx 0$ and $h \approx 0$ are assumed in the following. Each permanent magnet synchronous type linear motor in this system is connected to a digital drive. The ratios of force constants of motors have been priorly measured offline with $k_{mx} = 0.45$ and $k_{my} = 1.05$. The linear positions of one X-axis motor (i.e., x) and two Y-axis motors (i.e., y_1 and y_2) are directly measured by linear encoders with a resolution of 0.5 μm after quadrature. Thus, during implementation, the signals of y and α can be obtained through the following equations after necessary calibration:

$$y = \frac{y_1 + y_2}{2}, \quad \alpha = \frac{1}{l_e}(y_2 - y_1), \tag{46}$$

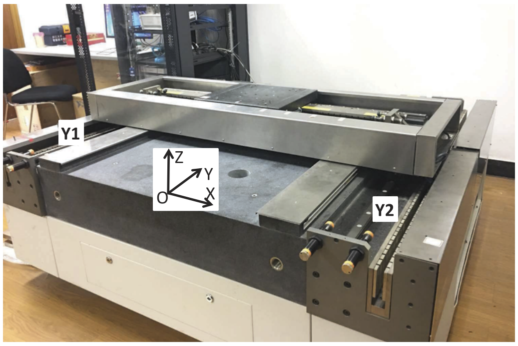

FIG. 4. Dual-driven H gantry.

which leads to an equivalent resolution of 0.476 μrad for α. The velocity signal is then computed by the difference of two consecutive position measurements. A dSPACE controller board is connected to the DLMD gantry. The control algorithms are directly compiled in MATLAB/Simulink and downloaded to dSPACE for implementation, and the sampling period is set as $T_s = 0.1$ ms.

B. Performance index notations

Same as that in 12, the following indices are used to qualify the performance:

$\bullet_M = \max_t\{|\bullet|\}$, the maximum absolute value of $\bullet$;

$\bullet_F = \max_{T-25\le t\le T}\{|\bullet|\}$, the maximum absolute value of $\bullet$ during the last 25 s; and

$L_2[u_y] = \left(\frac{1}{25}\int_{T-25}^{T}(|u_1|+|u_2|)^2 dt\right)^{1/2}$, the root-mean-square (rms) value of control efforts during the last 25 s.

C. Experimental results

To simulate the dynamic load condition in practice, both X and Y axes are commanded to move together along a "V" shape contour in the world frame OXY, as shown in Fig. 5. The corresponding desired smooth trajectories of two running periods are given in Fig. 6. As seen from Fig. 6, $y_d(t)$ is a unidirectional point-to-point (p2p) S-curve, while $x_d(t)$ is a bidirectional one.

To clearly illustrate the coupling effects under the dynamic load condition and simultaneously verify the effectiveness and performance improvement of the proposed synchronization method, the following three controllers are compared:

- **C1**: The proposed TITO precision synchronization controller in Sec. III and designed via DCDIARC. Λ is set as Λ = diag[240, 200, 150]. For simplicity, the robust term (25) is implemented as $v_s = -K_s s - K_e e - K_a\|e\|^2 s$, with $K_s = K_{s1} + K_{s2}$ and K_{s2} being a large enough constant gain matrix. Then, according to the tuning rules given in Ref. 31, the feedback gains are set as K_s = diag[60, 340, 100], K_e = diag[1000, 2000, 2000], and K_a = diag[1500, 3000, 3000]. The upper bound of $\hat{d}_c$ is set as $d_{c\ \max} = [1,1,2]^T$, with corresponding adaptation rates being $\gamma_d = diag[3000, 6000, 1500]$. In practice, system identification has been performed first to verify the previously proposed dynamical model, and the efficient information of system under normal conditions was obtained. Then, according to the offline identification results and to show the fast and accurate online estimation ability of DCDIARC, the initial values of parameter estimates are chosen as $\hat{\theta}(0) = [0.11, 0.14, 0.04, 0, 0.8, 1, 0.1, 0, 0.2, 8, 0, 0, 0]^T$, with the lower and upper bounds set as $\theta_{\min}$ = [0.08,0.05,0.01, −1,0.7,0.2,0.1,−1,0.2,5,−1,−0.5,−1]^T and $\theta_{\max}$ = [0.6, 0.4,0.3, 1,2,5,1,1,0.5,15,1,0.5,1]^T. The initial values of the corresponding adaptation rates are set as $\Gamma(0) = diag$[10, 30, 30, 10, 10, 30, 30, 10, 10, 10, 10, 10, 10], with related coefficients being $\rho_M = 5000$, $\mu = 0.02$, and $\kappa = 0.1$.
- **C2**: The static thrust allocation based synchronization control method proposed in Ref. 10 and also designed via DCDIARC. In this controller, the motor force of the X-axis, i.e., u_x, and the combined force of the Y-axis, i.e., $u_1 + k_{my}u_2$, are designed according to the individual linear motion dynamics of each axis first. Then, the thrusts of two parallel motors of the Y-axis are properly allocated according to the following relationship to keep the rotational moment balance of the crossbeam,

$$F_{m2}l_{m2} - F_{m1}l_{m1} + F_{r1}l_{g1} - F_{r2}l_{g2} = 0. \tag{47}$$

Meanwhile, considering the relatively smaller values of friction terms in condition (47) when compared with the electromagnetic forces in most practical applications, it can be simply implemented by

$$F_{m2}l_{m2} - F_{m1}l_{m1} \approx 0,\ i.e., k_{my}u_2 l_{m2} - u_1 l_{m1} \approx 0. \tag{48}$$

Then, for static thrust allocation, the coefficients l_{m1} and l_{m2} are set to be constant according to the initial position of the working head, i.e., $l_{m1}(x(0))$ and $l_{m2}(x(0))$, which will be priorly identified offline. Besides, the corresponding gains of

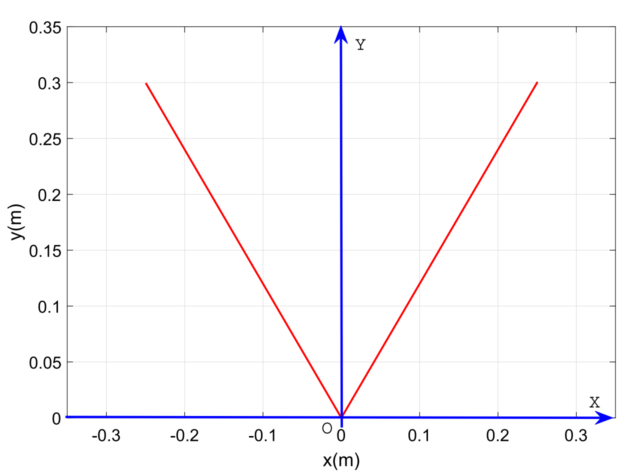

FIG. 5. "V" shape contour.

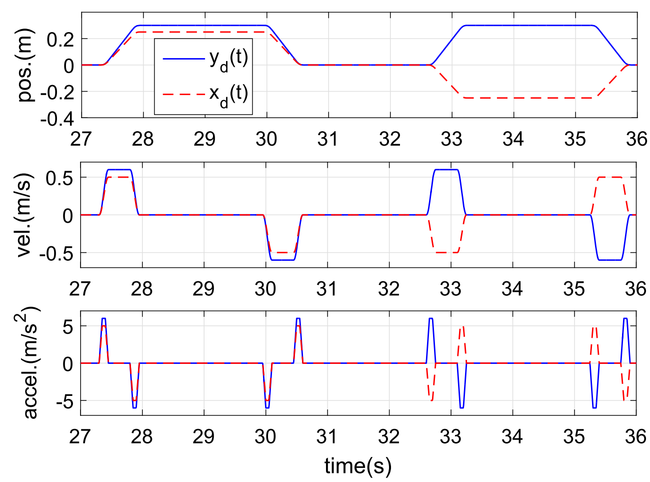

FIG. 6. Desired trajectories of X- and Y-axes.

the tracking controllers of X- and Y-axes are set to be the same as C1 for fair comparison.

- **C3**: The dynamic thrust allocation based synchronization control method proposed in Ref. 17 and also designed via DCDIARC. The controller structure is almost the same as C2. The only difference is the thrust allocation coefficients $l_{m1}(x)$ and $l_{m2}(x)$ are online-regulated according to the following equations to partly compensate the dynamic load effects,

$$l_{m2}(x) = \frac{l_m}{2} - \frac{M_x}{M_y}x, \quad l_{m1}(x) = \frac{l_m}{2} + \frac{M_x}{M_y}x. \tag{49}$$

Remark 1. Both thrust allocation based synchronization controllers C2 and C3 actually have no closed-loop feedback control of rotational dynamics. The reason for such a handling is that the rotational dynamics can be naturally stabilized by the stiffness of bearing balls, which can be taken as the inherent feedback gain of this dynamics. Thus, the transient performance of regulation of internal forces might not be actively guaranteed, which partly depends on the level of K_α. On the other hand, the proposed C1 aims to actively control the rotational dynamics for a better performance.

Furthermore, the following experimental sets are designed:

Set1: Experiments are performed under normal conditions, i.e., without payload, in which $l_{m1}(x(0)) \approx l_{m2}(x(0)) \approx 0.73$ m.

Set2: A 40 kg payload is added to the working head to test the robust performance of DCDIARC to parameter variations.

The comparative experimental results are plotted in Figs. 7–10. The corresponding performance indices are calculated and given in Table II.

First, the rotation angles shown in Fig. 9 clearly shows that C2 achieves the worst synchronization performance, i.e., significant rotational oscillations appear during moments A and B. As can be seen from Table II, the maximum oscillated amplitude of the rotational angle is 34.05 μrad. Simultaneously, additional control efforts need to be taken to deal with the subsequent excessive internal forces. Thus, under the same condition as in **set1**, corresponding oscillations occur on the control inputs of C2 among three controllers, as shown in Fig. 10, leading to the maximum control effort $L_2[u_y]$, as given in Table II. Thus, the direct relationship between the rotational angles and internal forces is clearly indicated. Meanwhile, the dynamic load effects have also been verified. Due to the movement of the working head, high-level synchronization between two parallel motors of the Y-axis obviously cannot be achieved by static thrust allocation. Especially, due to the complicated coupling, the tracking performance of C2 is also severely affected. As can be

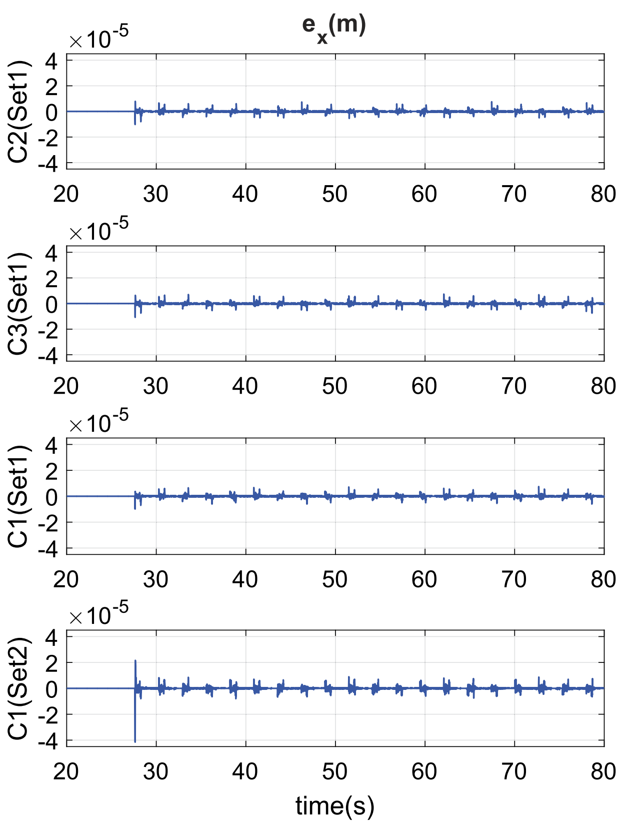

FIG. 7. Tracking errors of *x*.

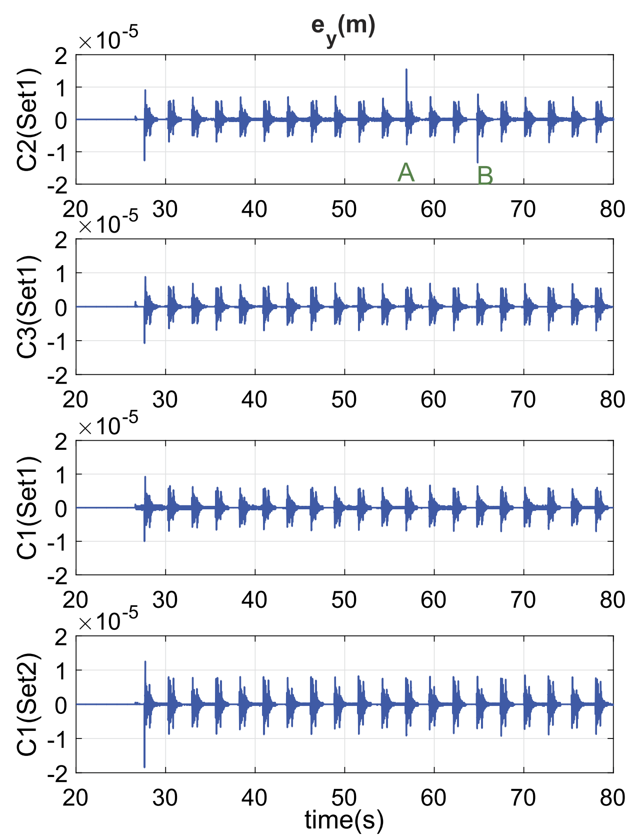

FIG. 8. Tracking errors of *y*.

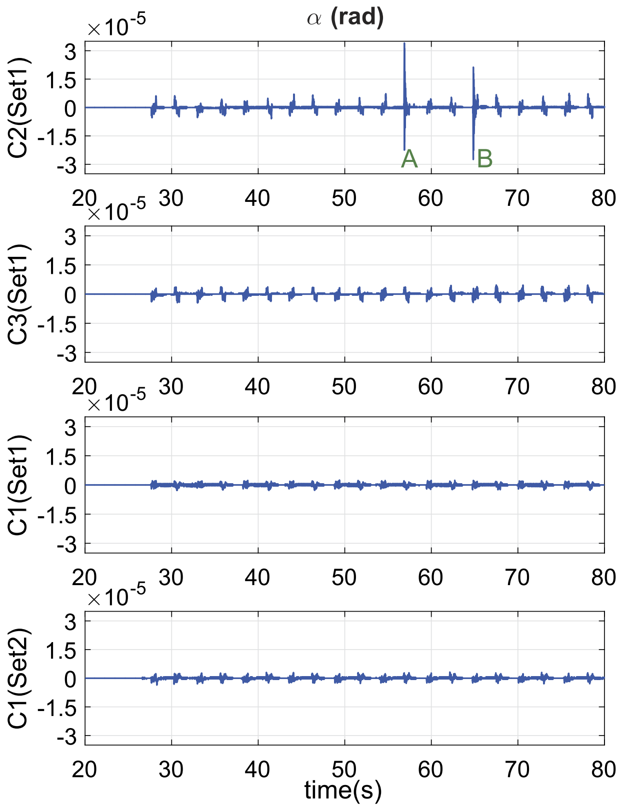

FIG. 9. Rotational angles of the crossbeam.

TABLE II. Performance index.

Indices	e_{xM}	e_{xF}	e_{yM}	e_{yF}	α_M	α_F	$L_2[u_y]$
Unit		μm			μrad		V
$C2(set1)$	10.2	7.52	15.5	15.5	34.05	34.05	2.124
$C3(set1)$	10.5	7.49	10.7	7.14	4.71	4.67	2.104
$C1(set1)$	9.9	7.46	10	7.08	2.86	2.76	2.102
$C1(set2)$	41.5	8.68	18.5	9.21	3.58	3.21	2.543

seen from Fig. 8, corresponding oscillated tracking errors have been caused during moments A and B.

On the other hand, C3 can achieve better synchronization performance than C2 due to the online computation of thrust allocation coefficients according to the feedback signal of x. As seen from Fig. 9, the rotational angles of C3 can be suppressed within a small level to regulate the internal forces efficiently during the whole running periods. Correspondingly, as seen from Figs. 7, 8, and 10, the oscillations appearing when controlled by C2 have been eliminated by C3 to achieve better tracking performance and smoother control inputs. Although the dynamic load effects have been partly compensated by C3 through the dynamic thrust allocation technique, the transient performance of the rotational dynamics and regulation of internal forces cannot be fully guaranteed due to the open-loop treatment of rotational dynamics.

Based on the above-mentioned analyses, it can be predicted that the proposed controller C1 could achieve best synchronization performance due to the direct control of rotational dynamics. As seen

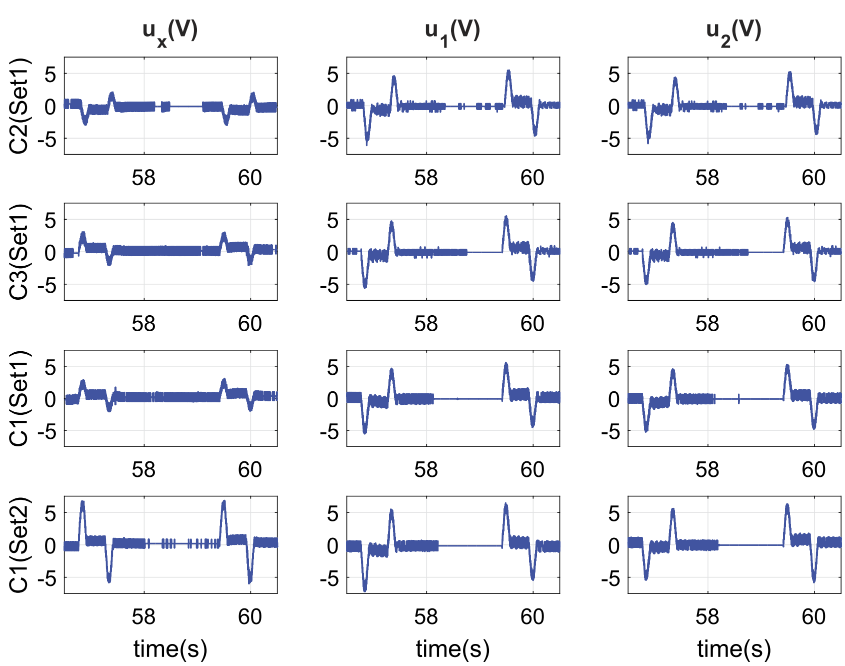

FIG. 10. Control inputs.

from Fig. 9, the rotational angles of C1 are highly suppressed during the whole history. Indicated by the values of α_F given in Table II, i.e., 2.76 μrad for C1 and 4.67 μrad for C3, the improvement in synchronization performance achieved by C1 is fully verified. Combined with the smallest control effort $L_2[u_y]$ of C1 among three controllers, it can be concluded that C1 can obtain best internal force regulation. Therefore, the best tracking performance is also achieved by C1 among three controllers, with the final tracking accuracy represented by e_{xF} = 7.46 μm and e_{yF} = 7.48 μm under **set1**, which is much better than that of C2 and clearly reflects the importance of regulation of internal forces to guarantee high-accuracy tracking of DLMD gantry systems.

On the other hand, combining the parameter estimation histories in Fig. 11 with those in Figs. 7 and 8, it can be seen that the initial tracking error and rotational angle are always large due to the poor model compensation at the beginning. However, owing to the fast and accurate online adaptation ability of DCDIARC, even under the heavy load condition, i.e., **set2**, the parameter estimates can converge quickly to provide better model compensation for effectively dealing with parametric uncertainties and dynamic coupling effects. Therefore, C1 can still maintain excellent control performance under **set2**, i.e., e_{xF} = 8.68 μm, e_{yF} = 9.21 μm, and α_F = 3.21 μrad, which is very close to the results of C1 under **set1**. Thus, the effectiveness and superiority of the proposed method have been completely verified. With the use of accurate online adaptation and active control of the rotation dynamics to deal with dynamic load effects, both high-accuracy tracking of two axes and high-level synchronization between two parallel motors, i.e., strong suppression of the rotational angles to avoid large internal forces, can be simultaneously achieved.

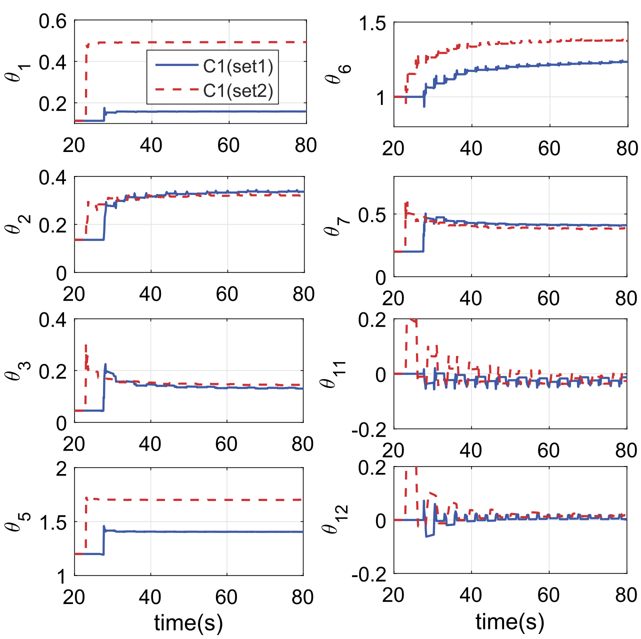

FIG. 11. Parameter estimation history.

V. CONCLUSION

In this paper, an accurate 3DOF mathematic model for a general DLMD gantry with dynamic load is presented, where both rigid and flexible dynamics are considered, i.e., the linear motions of two axes and the rotation of the crossbeam. The presented model clearly shows the generation of the rotational motion along with excessive internal forces, the dynamic load effects, and the complicated coupling properties in this system. Built upon this knowledge, a novel TITO synchronization control method is then proposed by directly controlling the linear dynamics for high-accuracy tracking of both axes and actively suppressing the rotational dynamics to regulate the internal forces for high-level synchronization between parallel motors. Meanwhile, the DCDIARC algorithm is utilized to account for the parametric uncertainties and uncertain nonlinearities in practical applications and compensate the dynamic load effects. The systematic design procedure of the controller with detailed stability proof has been clearly given. Comparative experimental results show that both excellent tracking performance of the linear motions of X- and Y-axes and high suppression ability of the rotational angle during both transient and steady-state periods can be achieved by the proposed synchronization method, indicating the effectiveness of this approach and its superiority over existing approaches.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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